**Controlled birds are a thing**
Engineers created a robot with the feet a peregrine falcon that can perch and carry goods like a bird. **https://dgit.in/dec21-32**

**Knotty mathematics**
DeepMind's AI uses machine learning to assist disentangle the mathematics of knots. **https://dgit.in/dec21-33**



1. First click on the video recorder symbol in the left toolbar.
2. Now you will be prompted with a variety of selection from preset videos. You can choose one from these presets or even use your PC's camera by selecting your active camera from the options just above these presets.

## ADDING YOUR OWN OBJECTS AND ASSETS INTO THE SCENE

To add your objects into the project, you can use the Add button on the menu bar, or the Add Object button, just like you can with other objects. You can also use the Assets panel's Add Assets button. When you select this option, an object will be added to your project, but not to the scene. This is where you'll see a list of all the objects and assets you've added to a project. To add these to the scene just drag these objects to the scene panel. This will create an instance of the object into the scene panel.

You will also see a few other options when you use this button from the assets panel. You can use this to create materials, scripts and playback controllers for playing animation and even add audio/music in your effects.

To add a 3D Object into the project:

1. Select the 'Import from Computer' option.
2. Locate the sample content folder on your computer, and select it.

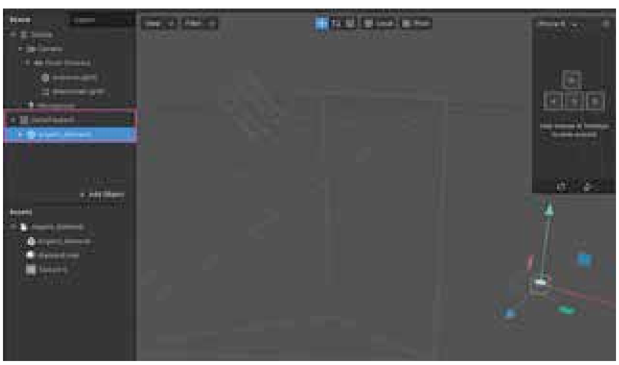
The object will also appear in the Viewport once you have placed its instance in the scene panel.

## EDITING OBJECTS AND ASSETS IN YOUR PROJECT

When an object or asset is selected in the Scene/Assets panel, you can use the Inspector tab on the right, to change its properties:

From here, you can:

- Change which layer it's currently on.
- Adjust whether or not it's visible in the effect.



- Change its position, scale and rotation.
- Create or apply a material to the mesh.

## CHANGING SCALE, ROTATION AND POSITION

You can change these properties of an object easily by using the manipulators located at the top-center of the Viewport. They're a super useful and quick way to experiment with your 3D objects and each of them can be used for:

- Position – to move an object around.
- Rotation – to rotate an object.
- Scale – to make changes to the object size..

## USING THE PHONE SIMULATOR

Let's take a quick look at the Simulator which represents a device's screen. It can be used to preview how your effect will look on different types of device.

To change the size of the Simulator, click and drag any of its corners or sides. Click on the top of Simulator to change the device type from the dropdown options. You may also use the Simulator to modify the rotation of the gadget and switch between the front and back camera in a device. The default setting for the Simulator will be set to Simulator Orbit, which is a representation of what the camera can view. In this, you can adjust the camera and alter the view by clicking and dragging your mouse. Another option, the Simulate Touch, simulates a user touching or tapping the device's screen, which is handy for testing effects that include interaction with the screen.

## TEXTURES AND MATERIALS

In advance tutorials you will be using textures and materials a lot to build the desirable appearance of many objects in your scene. It's listed in the Assets panel and can be selected to view its properties in the Inspector tab on your right. Starting with the Shader Type, there are several different Shader Types to choose from. Which one you'll use depends on the effect you're trying to produce. For beginners, the Standard shader is quite good enough for making 3D objects look realistic because of the way it mimics light on an object.

There are also other fields that you can adjust from the options here to further customise the objects.

## THE PATCH EDITOR AND CONSOLE

To open the scripting console, use the button at the top of the toolbar under the view option.

This button can also be used to access the Patch Editor, which is the visual programming environment for the Spark AR Studio. You can use it to further add interaction, logic, and animation to your effects. It can also be used to generate new textures and make complicated material changes.

## PREVIEWING YOUR EFFECTS

You can see how your effect looks and performs on an actual device by using the Spark AR Player app. It's available on the App Store and Play Store. Once you've connected your device to your computer using a USB cable, you'll be able to click on this button and select the 'Send To Connected Device' option.

Using this, you will see the effect working in the Spark AR Player app. Or, you can select the 'Send to App' option